UniGD: A Unified Generative-Discriminative Framework for Industrial Retrieval

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Abstract

Generative retrieval (GR) is a promising paradigm for industrial search advertising, yet its deployment is constrained by strict relevance and latency requirements. Existing systems cascade GR with an independent relevance model, decoupling the generative likelihood objective from query–ad relevance discrimination, which compromises effectiveness and increases serving costs. We propose a Unified Generative-Discriminative framework (UniGD) that integrates retrieval and relevance scoring within a single model. To mitigate gradient interference in joint optimization, UniGD introduces Conflict-Aware Gradient Enhancement (CAGE) to adaptively coordinate the two objectives. UniGD further designs a Codebook-Anchored Representation Module (CAM) that anchors item representations to frozen hierarchical codebooks distilled from a multimodal pretrained model, thereby endowing them with rich and generalizable semantic priors. For heterogeneous short-video, product, and live-stream ads, UniGD proposes Heterogeneous Ad-material Modeling (HAM), which captures cross-type semantic commonality over a shared backbone while preserving type-specific modeling capacity. Online A/B tests on Kuaishou search advertising platform show that UniGD raises ad revenue by 5.78%, reduces inference latency by 33.1%, and improves discriminative relevance estimation. On NQ320K and MS300K, UniGD improves Recall@10 over the strongest reproduced GR baseline by 8.44% and 3.19%, respectively.

Keywords: Generative retrieval, Search advertising

Introduction

Search advertising systems must retrieve semantically relevant ads from hundreds of millions of candidates under stringent latency constraints. Conventional multi-stage retrieval pipelines tackle this challenge through progressive candidate filtering, but each stage is typically optimized independently, leading to inconsistent objectives and limited overall retrieval effectiveness. Generative Retrieval (GR) (Tay et al. 2022; Bevilacqua et al. 2022) has recently attracted increasing attention as a promising paradigm for large-scale candidate retrieval. Through autoregressive decoding, GR casts candidate retrieval as the conditional generation of Semantic Identifiers (SIDs) (van den Oord, Vinyals, and Kavukcuoglu 2017; Rajput et al. 2023; Deng et al. 2025). This generative paradigm moves beyond conventional matching-based retrieval and enables unified modeling of query-candidate semantic interactions.

In real-world industrial search advertising deployments, GR is typically cascaded with a downstream relevance model to ensure the relevance of the final results. Specifically, GR first retrieves top-K candidate ads by autoregressive generation likelihood, and the relevance model then assigns discriminative relevance scores using query-ad matching signals. Despite its practical adoption, this cascaded paradigm still faces several key challenges in industrial deployment.

(1) **Objective discrepancy and serving overhead.** The discrepancy between generative retrieval and discriminative relevance scoring may cause relevant ads with low generation probabilities to be prematurely discarded during top-K truncation, before reaching the downstream relevance model for rescoring (Tang et al. 2024; Cheng et al. 2025). Moreover, the cascaded GR paradigm introduces additional online inference latency. Although previous studies introduce ranking supervision to reshape the relative ordering of generation likelihoods (Zhou et al. 2024; Li et al. 2024b), such preference signals do not directly produce explicit pointwise relevance scores. Consequently, they are unsuitable for threshold-based filtering and offer limited reliability as relevance features for downstream modules.

(2) **Representations for long-tail and new ads.** Reliable relevance scoring requires discriminative ad-side representations, but long-tail and new ads often lack sufficient interaction signals for representation learning. This issue is particularly pronounced on our platform, where new ads account for as much as 36.9% of daily traffic. Existing cascaded or auxiliary discriminative modules (Tang et al. 2024; Dai et al. 2025) typically rely on separately trained ad encoders or periodically updated ad embeddings, making it difficult to provide reliable representations immediately after indexing.

(3) **Unified modeling of heterogeneous ad materials.** Industrial search advertising covers multiple ad material types, including short-video, product, and live-stream ads, which exhibit significant semantic heterogeneity. Existing GR methods lack explicit modeling of cross-material differences, and thus struggle to accommodate the semantic characteristics of different material types within a shared SID semantic space, limiting the generalization ability of GR in

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industrial advertising deployments.

To address these challenges, we propose **UniGD**, a **Uni**fied **G**enerative-**D**iscriminative framework that integrates generative retrieval and relevance discrimination within a single model. Unlike cascaded GR, UniGD jointly models autoregressive generation and query-ad relevance discrimination over a shared backbone, achieving mutual reinforcement and optimization synergy while eliminating the serving overhead of an additional relevance model. Specifically, UniGD introduces three key designs. (1) **Conflict-Aware Gradient Enhancement (CAGE).** Since jointly optimizing generative and discriminative objectives over a shared backbone may produce inconsistent gradient update directions, we introduce CAGE to coordinate their gradients during backpropagation, enabling the two objectives to benefit from joint optimization. (2) **Codebook-Anchored Representation Module (CAM).** To address the cold-start bottleneck in ad retrieval under sparse behavioral supervision, we design CAM, which anchors item-side representations to frozen hierarchical codebooks derived from a large-scale multimodal pretrained model, allowing long-tail and new ads to inherit its semantic priors and obtain reliable representations immediately upon indexing. (3) **Heterogeneous Ad-material Modeling (HAM).** Considering the semantic heterogeneity across ad materials, we propose HAM to assign each material type to a dedicated semantic space and inject material-aware signals into the unified backbone. In this way, both SID generation and relevance discrimination can focus on the appropriate signal sources for each material type, thereby enhancing the unified generative-discriminative process while avoiding an over-simplified homogeneous formulation.

We validate UniGD through both industrial-scale online deployment and public benchmark evaluation. Online A/B tests on Kuaishou search advertising platform demonstrate that UniGD improves ad revenue by 5.78% while reducing inference latency by 33.1%, from 13.00 ms to 8.70 ms. On the NQ320K and MS300K benchmarks, UniGD outperforms the strongest reproduced GR baseline in Recall@10 by 8.44% and 3.19%, respectively. The main contributions of this paper are summarized as follows:

- We propose UniGD, a unified generative-discriminative retrieval framework that jointly performs autoregressive SID generation and explicit discriminative relevance scoring within a single shared model, eliminating the need for a cascaded relevance model.
- We develop three key components for effective industrial deployment: CAGE mitigates optimization conflicts between the generative and discriminative objectives; CAM provides reliable item-side representations for long-tail and newly indexed ads; and HAM captures semantic heterogeneity across short-video, product, and live-stream ads.
- We conduct extensive online and offline evaluations, demonstrating that UniGD substantially improves retrieval effectiveness and advertising revenue while reducing online inference latency.

Related Work

Generative Methods

Unlike traditional term-based (Robertson et al. 1995; Nogueira and Lin 2019; Raffel et al. 2020) and dense retrievers (Karpukhin et al. 2020; Xiong et al. 2021; Zhan et al. 2020; Ni et al. 2022; Qu et al. 2021), generative retrieval recasts large-scale candidate retrieval as sequence generation. Early studies established the technical foundation via document-identifier generation, query-generation augmentation, prefix-aware decoding, and learnable tokenization (Tay et al. 2022; Zhuang et al. 2022; Wang et al. 2022; Sun et al. 2023). Subsequent work advanced the paradigm through scalable multi-stage training, efficient simultaneous decoding, and semantic docid design (Zeng et al. 2024; Zeng, Luo, and Zamani 2024; Zhou et al. 2022; Li et al. 2024a; Guo et al. 2025), and extended SID-based generation to sequential recommendation and industrial recommendation and advertising pipelines (Rajput et al. 2023; Deng et al. 2025; Zheng et al. 2025; Xue et al. 2026). Despite this progress, existing GR methods largely optimize generation probability via MLE and rely on beam search at inference, which inflates high-frequency SIDs and underestimates semantically relevant long-tail and new items.

Generative-Discriminative Methods

Several studies integrate ranking-oriented or discriminative signals into generative retrieval and recommendation (Zhang et al. 2025; Sun et al. 2026; Fu et al. 2026; Chen et al. 2025). Representative efforts introduce ranking-oriented objectives or document-level relevance optimization into MLE-based GR (Zhou et al. 2024; Li et al. 2024b; Mekonnen, Tang, and de Rijke 2025), align generation with query-item relevance through end-to-end training (Pang et al. 2025), and design discriminative identifiers to sharpen candidate distinction (Cheng et al. 2025). While these works make valuable progress in bridging generation with relevance-oriented ranking, search advertising's rapid ad turnover demands higher retrieval and discrimination accuracy, calling for a unified framework that tightly couples candidate generation with relevance estimation while ensuring scalability.

Methodology

Overview

In this paper, we propose UniGD, a unified decoder-only framework that integrates SID-based candidate generation and query-ad relevance scoring. As illustrated in Figure 1, HAM captures material-specific semantics for both generative and discriminative learning, CAM derives codebook-anchored ad representations from frozen SID codebooks for relevance scoring, and CAGE coordinates the two objectives during joint optimization. Figure 2 presents the corresponding industrial serving architecture, where candidate generation and relevance scoring are performed within the same model, eliminating the external online relevance model and thereby reducing inference latency and serving overhead.

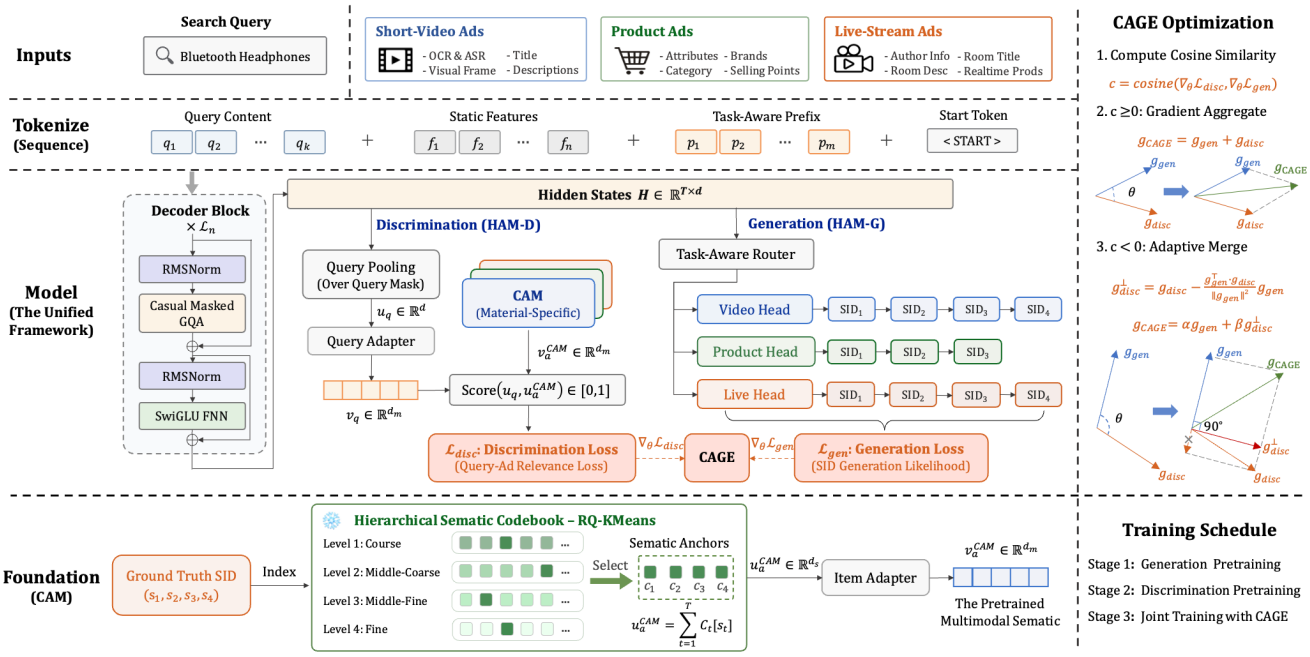


Figure 1: Training framework of UniGD. HAM-G and HAM-D jointly optimize the generative and CAM-based discrimination, while CAGE coordinates their gradients through a three-stage training scheme.

Unified Generative-Discriminative Modeling

Given a query q and an ad corpus $\mathcal{C}$, UniGD unifies autoregressive SID generation and discriminative relevance estimation within a shared decoder-only backbone. It jointly models the conditional generation probability $p(\mathbf{s}(a)|q)$ and the explicit relevance score $r(q,a)$.

SID tokenization. Each ad $a \in \mathcal{C}$ is indexed by a hierarchical semantic identifier (SID) $\mathbf{s}(a) = (s_1(a), \ldots, s_T(a))$ constructed using RQ-KMeans. For autoregressive generation, we augment the model vocabulary with level-specific SID tokens and serialize $\mathbf{s}(a)$ as a token sequence, where $s_t(a) \in \mathcal{V}_t$ denotes the SID token at level t. The same SID indices are used by CAM to retrieve the corresponding codewords for relevance estimation.

Autoregressive SID generation. Conditioned on a query q, UniGD factorizes the SID generation probability as

$$p(\mathbf{s}(a) \mid q) = \prod_{t=1}^{T} p\left(s_t(a) \mid q, \mathbf{s}_{<t}(a)\right), \tag{1}$$

where $\mathbf{s}_{<t}(a)$ denotes the preceding SID prefix. At decoding step t, the shared decoder produces the last-layer hidden state $\mathbf{h}_t^{\text{sid}} \in \mathbb{R}^d$ and predicts the next SID token over $\mathcal{V}_t$ as

$$p(s_t(a) \mid q, \mathbf{s}_{<t}(a)) = \text{Softmax}\big(\mathbf{W}_t \mathbf{h}_t^{\text{sid}} + \mathbf{b}_t\big)_{s_t(a)}. \tag{2}$$

where $\mathbf{W}_t \in \mathbb{R}^{|\mathcal{V}_t| \times d}$ and $\mathbf{b}_t \in \mathbb{R}^{|\mathcal{V}_t|}$ are the prediction parameters associated with SID level t, which are further adapted by HAM to capture material-specific semantics. The generative objective for a training pair (q, a) is

$$\mathcal{L}_{\text{gen}} = -\sum_{t=1}^{T} \log p\left(s_t(a) \mid q, \mathbf{s}_{<t}(a)\right). \tag{3}$$

Discriminative relevance estimation. UniGD derives the query representation for relevance estimation from the query-token hidden states of the shared decoder. Let $\mathbf{H}_q \in \mathbb{R}^{N_q \times d}$ denote the last-layer hidden-state matrix at the N_q valid query-token positions. We aggregate these states into a query-level representation $\mathbf{u}_q = \text{Agg}(\mathbf{H}_q)$, where $\text{Agg}(\cdot)$ is instantiated as mean pooling in our implementation. For a candidate a, let $\mathbf{u}_a^{\text{CAM}} \in \mathbb{R}^{d_s}$ denote its codebook-anchored representation constructed by CAM. To align $\mathbf{u}_q$ and $\mathbf{u}_a^{\text{CAM}}$ in a shared semantic space, we employ a query-side adapter g_q and an ad-side adapter g_a, respectively. The relevance logit is then computed as

$$r(q,a) = \psi\big(\big[g_q(\mathbf{u}_q); g_a(\mathbf{u}_a^{\text{CAM}})\big]\big), \tag{4}$$

where $[\cdot;\cdot]$ denotes vector concatenation and ψ is a lightweight scoring network that outputs $r(q,a) \in [0,1]$.

Given a supervised candidate set $\mathcal{D}_q = \{(a_i, y_i)\}_{i=1}^{K}$ for query q, where $y_i \in [0,1]$ is the normalized relevance target derived from human annotation or an offline relevance model, the pointwise loss is

$$\mathcal{L}_{\text{point}} = \frac{1}{K} \sum_{i=1}^{K} \left(r(q,a_i) - y_i\right)^2. \tag{5}$$

To further enforce relative relevance ordering, we introduce an auxiliary pairwise ranking loss. Given a positive ad a^+ and M negative ads $\{a_j^-\}_{j=1}^{M}$ for query q, it is defined as

$$\mathcal{L}_{\text{pair}} = \frac{1}{M} \sum_{j=1}^{M} \max\left(0, \gamma - r(q,a^+) + r(q,a_j^-)\right), \tag{6}$$

where $\gamma \in (0,1)$ is the ranking margin. The complete discriminative objective combines pointwise score supervision and pairwise ranking:

$$\mathcal{L}_{\text{disc}} = \mathcal{L}_{\text{point}} + \lambda \mathcal{L}_{\text{pair}}, \tag{7}$$

where λ controls the contribution of pairwise supervision.

To fully develop both generative and discriminative capabilities, UniGD adopts a three-stage training scheme. Stage1 and Stage2 separately optimize $\mathcal{L}_{\text{gen}}$ and $\mathcal{L}_{\text{disc}}$ to initialize the corresponding task components. Stage3 jointly optimizes both objectives with CAGE coordinating their gradients over the shared parameters.

Codebook-Anchored Representation Module

Discriminative relevance estimation requires informative ad-side representations under strict online latency constraints, which is particularly challenging for new and long-tail ads with sparse interaction signals. To address this issue, we introduce the *Codebook-Anchored Representation Module (CAM)*. Given an ad's SID, CAM constructs a quantized approximation of its multimodal representation by looking up and combining the corresponding codewords.

Multimodal residual quantization. We employ a large-scale multimodal encoder, pretrained on a large corpus of ad creatives, to encode the heterogeneous content of each ad into an informative semantic representation:

$$\mathbf{v}_a = \text{Enc}_{\text{MM}}(a) \in \mathbb{R}^{d_s}. \tag{8}$$

RQ-KMeans is trained offline on the ad representations in the indexing corpus to learn T residual codebooks. Let $\mathbf{C}_t \in \mathbb{R}^{|\mathcal{V}_t| \times d_s}$ denote the codebook at level t. Given the residual $\boldsymbol{\rho}_{t-1}(a)$ left by the preceding levels, the SID token at level t is assigned to its nearest codeword:

$$s_t(a) = \arg\min_{k \in \mathcal{V}_t} \left\| \boldsymbol{\rho}_{t-1}(a) - \mathbf{C}_t[k] \right\|_2^2, \tag{9}$$

where $\boldsymbol{\rho}_0(a) = \mathbf{v}_a$. Earlier levels capture the dominant components of the multimodal representation, while subsequent levels quantize the remaining residuals to progressively encode finer-grained information.

Codebook-anchored reconstruction. CAM retrieves the codeword indexed by the SID token at each level, $\mathbf{c}_t(a) = \mathbf{C}_t[s_t(a)]$, and recursively updates the residual as $\boldsymbol{\rho}_t(a) = \boldsymbol{\rho}_{t-1}(a) - \mathbf{c}_t(a)$. The ad-side representation is reconstructed by summing the selected codewords:

$$\mathbf{u}_a^{\text{CAM}} = \sum_{t=1}^{T} \mathbf{c}_t(a). \tag{10}$$

Consequently, $\mathbf{v}_a = \mathbf{u}_a^{\text{CAM}} + \boldsymbol{\rho}_T(a)$, where $\boldsymbol{\rho}_T(a)$ denotes the final quantization residual. Thus, $\mathbf{u}_a^{\text{CAM}}$ provides a quantized reconstruction of the original multimodal representation and serves as the ad-side input for discriminative relevance estimation. The codebooks remain frozen throughout UniGD training, preserving the correspondence between SID tokens and their offline-learned codewords and thereby providing stable semantic anchors. Once an ad is assigned an SID, CAM requires only T codebook lookups and a lightweight vector summation, enabling relevance estimation immediately after indexing without online ad-side encoder inference or complex feature construction.

Heterogeneous Ad-material Modeling

Industrial search advertising contains heterogeneous ad materials, including short-video, product, and live-stream ads. They differ substantially in content modalities, structures, presentation formats, and information density, resulting in different semantic distributions and relevance patterns. A homogeneous modeling scheme may therefore obscure material-specific semantics. We introduce *Heterogeneous Ad-material Modeling (HAM)* to specialize codebook construction, SID generation, and relevance estimation by material type while retaining the shared backbone.

Let $m \in \mathcal{M} = \{\text{video}, \text{product}, \text{live}\}$ denote the material type. HAM performs residual quantization separately for each type, yielding material-specific codebooks $\{\mathbf{C}_t^{(m)}\}_{t=1}^{T_m}$ and SID vocabularies $\{\mathcal{V}_t^{(m)}\}_{t=1}^{T_m}$. For generation, it replaces the output parameters in Eq. 2 with $\mathbf{W}_t^{(m)}$ and $\mathbf{b}_t^{(m)}$, which predict over $\mathcal{V}_t^{(m)}$. During training, m is obtained from the target ad's material-type metadata. At inference, all material-specific heads perform type-specific beam search in parallel using the shared decoder hidden states, and the resulting typed SID candidates are merged for relevance scoring without additional backbone forward passes. For relevance estimation, CAM retrieves codewords from the codebooks of type m and reconstructs the material-aware ad representation

$$\mathbf{u}_a^{\text{CAM},(m)} = \sum_{t=1}^{T_m} \mathbf{C}_t^{(m)}[s_t(a)]. \tag{11}$$

Substituting $\mathbf{u}_a^{\text{CAM},(m)}$ in the relevance scoring function yields the material-aware relevance score $r(q,a,m)$. In this way, generation and discrimination operate on consistent material-specific SID semantics while sharing the backbone and scoring architecture across materials.

Conflict-Aware Gradient Enhancement

Joint optimization may produce conflicting gradients from the generative and discriminative objectives over the shared decoder parameters, leading to optimization interference. We therefore introduce *Conflict-Aware Gradient Enhancement (CAGE)* to adaptively coordinate their gradients while prioritizing autoregressive SID generation.

Let θ_s denote the shared decoder parameters. We compute the task gradients $\mathbf{g}_{\text{gen}} = \nabla_{\theta_s} \mathcal{L}_{\text{gen}}$ and $\mathbf{g}_{\text{disc}} = \nabla_{\theta_s}(\eta \mathcal{L}_{\text{disc}})$, where $\eta > 0$ rescales the discriminative gradient to reduce the magnitude imbalance between the two objectives. Their directional agreement is measured by the cosine similarity $c = \cos(\mathbf{g}_{\text{gen}}, \mathbf{g}_{\text{disc}})$. When $c \geq 0$, the gradients are compatible and directly combined. When $c < 0$, CAGE removes the conflicting component of the discriminative gradient by projecting it onto the orthogonal complement of the generative direction:

$$\mathbf{g}_{\text{disc}}^{\perp} = \mathbf{g}_{\text{disc}} - \frac{\mathbf{g}_{\text{gen}}^{\top} \mathbf{g}_{\text{disc}}}{\|\mathbf{g}_{\text{gen}}\|_2^2} \mathbf{g}_{\text{gen}}. \tag{12}$$

The gradient for updating the shared decoder is

$$\mathbf{g}_{\text{CAGE}} = \begin{cases} \mathbf{g}_{\text{gen}} + \mathbf{g}_{\text{disc}}, & c \geq 0, \\ \alpha(c)\mathbf{g}_{\text{gen}} + \beta(c)\mathbf{g}_{\text{disc}}^{\perp}, & c < 0, \end{cases} \tag{13}$$

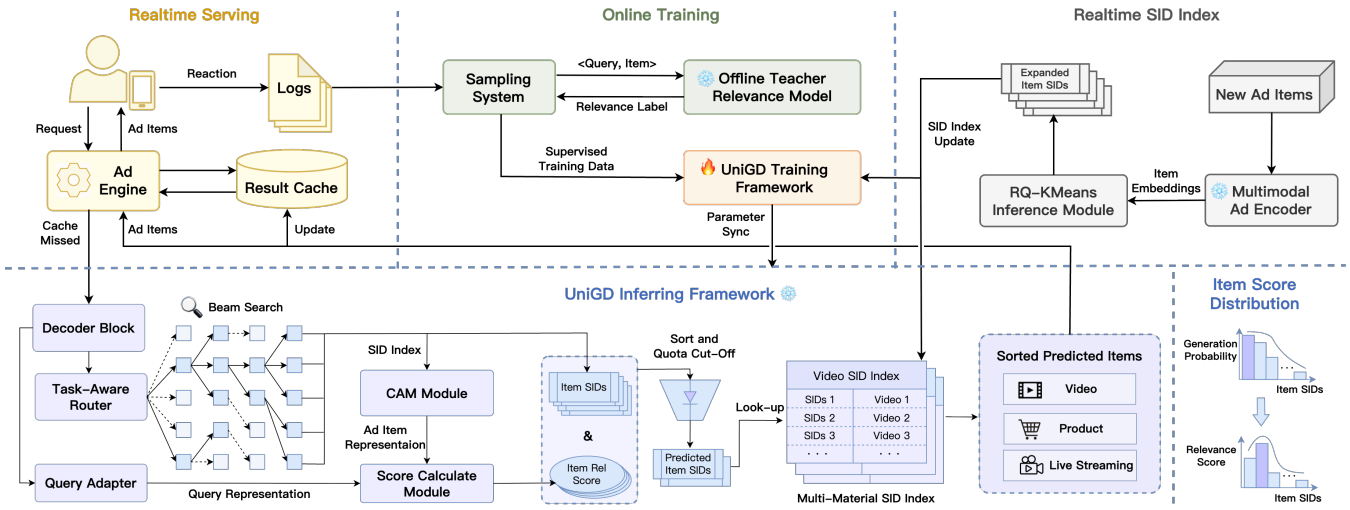


Figure 2: Online serving pipeline of UniGD in industrial search advertising.

where $\alpha(c) = 1 + \kappa(-c)$ and $\beta(c) = 1 - \kappa(-c)$, with $\kappa \in [0, 1]$ controlling the adaptation strength. As c decreases, CAGE increasingly favors the generative direction, preserving SID generation while retaining non-conflicting discriminative supervision. The resulting gradient is used to update the shared decoder, while task-specific parameters are optimized by their respective objectives.

Industrial Online Serving

Figure 2 illustrates UniGD's continual learning, incremental indexing, and real-time serving pipeline deployed in our search advertising system. Request contexts and user feedback are collected to form query-ad training samples, with fine-grained relevance targets provided by an offline teacher model. The trained parameters are periodically updated on the inference servers. Meanwhile, the real-time SID index maintains material-specific SID-ad mappings. New or updated ads are quantized into SIDs and incrementally indexed. Since these mappings are independent of model updates, newly indexed ads can enter retrieval without rebuilding a model-dependent embedding index.

During serving, UniGD generates candidate SIDs via beam search and derives the query representation from the shared decoder. CAM reconstructs each candidate's ad-side representation from its material-specific codebooks, and the lightweight scoring module reranks the candidates using explicit relevance scores. The retained SIDs are then resolved to actual ads through the real-time index. This unified pipeline removes the need for both a separately deployed online relevance model and per-candidate ad-side encoder inference.

Experiments

Experimental Settings

Datasets. We evaluate UniGD on two public generative retrieval benchmarks and one industrial search advertising dataset. NQ320K, derived from Natural Questions (Kwiatkowski et al. 2019), contains 320K training queries and 7,830 test queries over Wikipedia articles. MS300K (Zhou et al. 2022), curated from MS MARCO (Bajaj et al. 2016), contains 320K documents, 360K training queries, and 772 test queries. We follow the data splits used by DDRO (Mekonnen, Tang, and de Rijke 2025) for comparison with prior work. For public benchmarks, query-document pairs are labeled as relevant (positive) or non-relevant (negative). Our industrial dataset covers product, short-video, and live-stream ads collected from our production search advertising logs. It contains 153M positive pairs for generative training and 248M pairs for discriminative training. The test sets contain 277K pairs over 1K queries and over 10K human-labeled pairs, respectively.

Evaluation Metrics. For generative retrieval, we report Recall@1/5/10 and MRR@10, following prior work (Wang et al. 2022; Zhou, Dou, and Wen 2023; Zhou et al. 2024, 2022). On public benchmarks, significance is assessed against the strongest reproduced baseline using a two-tailed paired t-test over per-query values ($p < 0.05$). For relevance estimation on our industrial dataset, we report AUC, Spearman and Pearson to evaluate binary discrimination, rank consistency and score correlation. Relevance labels range from 0 (irrelevant) to 3 (highly relevant); for AUC, labels 0-1 are treated as negative and 2-3 as positive.

Baselines. For the industrial evaluation, we compare UniGD with the production cascaded system, which performs generative retrieval followed by an independently deployed relevance model. On the public benchmarks, we compare UniGD with representative generative retrieval methods. DSI (Tay et al. 2022) formulates retrieval as document-identifier generation; NCI (Wang et al. 2022) uses semantically structured numeric identifiers; Ultron (Zhou et al. 2022) combines keyword and semantic document identifiers with multi-stage training; LTRGR (Li et al. 2024b) incorporates ranking supervision into generative retrieval, GenRRL (Zhou, Dou, and Wen 2023) optimizes generation using relevance feedback, and DDRO (Mekonnen, Tang, and de Rijke 2025) combines PQ-based identifiers with direct

Model	NQ320K				MS300K			
	Recall@1	Recall@5	Recall@10	MRR@10	Recall@1	Recall@5	Recall@10	MRR@10
DSI (SI)	27.42	47.26	56.58	34.31	25.74	43.58	53.84	33.92
NCI (SI)	32.69	55.82	69.20	42.84	29.54	57.99	67.28	40.46
Ultron (TU)	33.78	54.20	67.05	42.51	29.82	60.39	68.31	42.53
LTRGR (SI)	32.80	56.20	68.74	44.80	32.69	64.37	72.43	47.85
GenRRL (Sum)*	35.82	57.04	<u>71.49</u>	45.93	32.78	63.91	<u>75.80</u>	45.98
DDRO (TU)*	40.52	52.81	55.63	45.68	<u>37.92</u>	<u>66.13</u>	73.72	<u>49.74</u>
DDRO (PQ)*	<u>48.57</u>	<u>63.78</u>	66.94	<u>55.18</u>	32.61	64.05	72.71	45.43
UniGD (Ours)	**72.33**†	**75.84**†	**77.52**†	**74.21**†	**60.64**†	**71.85**†	**78.22**†	**65.12**†

Table 1: Generative retrieval performance (%) on NQ320K and MS300K with representative generative retrieval baselines. **Bold**: best; <u>underlined</u>: second best. $*$ denotes results reproduced under our unified protocol. † indicates a statistically significant improvement over the strongest reproduced baseline for each metric under a two-tailed paired t-test over per-query metric values ($p < 0.05$). SI, TU, and PQ denote Semantic ID, Title and URL, and Product Quantization, respectively.

preference optimization (Rafailov et al. 2023).

Implementation Details. For the public benchmarks, we instantiate UniGD with T5-base (220M parameters) and adopt the PQ-based SIDs of DDRO with $T = 24$. We reuse the frozen PQ centroids from DDRO as CAM codebooks. For our industrial setting, we use a six-layer decoder-only Transformer with approximately 60M parameters, while the discriminative components introduce only 1.12M additional trainable parameters. Our industrial SIDs use three quantization levels for product ads and four for short-video and live-stream ads. Across both settings, UniGD is trained with AdamW for two epochs in each stage. Stage I optimizes $\mathcal{L}_{\text{gen}}$ with a learning rate of 1×10^{-4} and a batch size of 256. Stage II freezes the shared decoder and optimizes only the query-side and item-side adapters with $\mathcal{L}_{\text{disc}}$, using a learning rate of 1×10^{-3}, $\lambda = 0.5$, and $\gamma = 0.2$. Stage III jointly optimizes both objectives using CAGE, with $\eta = 100$ and $\kappa = 0.5$, where η balances their gradient magnitudes. The learning rates are 5×10^{-5} for the shared decoder and 1×10^{-4} for the task-specific modules. Industrial training takes three days on eight A800 GPUs, with beam widths of 100 and 32 for public and industrial inference, respectively.

Public Benchmark Evaluation

Table 1 compares UniGD with representative generative retrieval methods on NQ320K and MS300K. UniGD consistently achieves the best performance across metrics, with statistically significant improvements over the strongest reproduced baselines. UniGD improves Recall@1 by more than 22 absolute points on both datasets, while yielding relative Recall@10 gains of 8.44% on NQ320K and 3.19% on MS300K. The improvements in Recall@1 and MRR@10 indicate that UniGD more reliably places relevant candidates at the top of the ranking, highlighting the benefit of explicit discriminative supervision for learning fine-grained relevance signals between queries and candidates. Meanwhile, the consistent gains in Recall@10 suggest that the benefit extends beyond top-rank refinement to improved candidate coverage through the shared generative backbone. The results on both benchmarks demonstrate the effectiveness and generalizability of unified generative and discriminative modeling.

Variant	Recall@1	Recall@5	Recall@10	MRR@10
Our industrial test set				
UniGD	**19.57**	**32.83**	**53.37**	**27.20**
w/o CAM	13.24	26.67	46.33	20.21
w/o HAM	18.86	32.13	52.43	26.39
w/o CAGE	18.36	31.23	51.57	25.62
Baseline	12.53	25.82	43.98	19.14
NQ320K				
UniGD	**72.33**	**75.84**	**77.52**	**74.21**
w/o CAM	45.52	58.96	65.26	52.03
w/o CAGE	67.22	71.45	74.38	69.18
Baseline	42.50	56.63	61.18	48.62

Table 2: Comparison and ablation results on our industrial test set and NQ320K.

Industrial Offline Evaluation

We evaluate UniGD on our industrial dataset from two complementary perspectives: generative retrieval and discriminative relevance estimation.

Generative retrieval. Table 2 compares UniGD with a GR baseline using the same backbone and SID configuration but trained only for autoregressive SID generation. On our industrial test set, UniGD improves Recall@1 and MRR@10 by 7.04 and 8.06 points, respectively, demonstrating the benefit of jointly modeling candidate generation and explicit discriminative relevance estimation. Through the shared decoder, the discriminative objective complements token-level generation with relevance supervision, encouraging query representations to capture informative query–candidate matching signals.

Relevance estimation. We evaluate relevance prediction on our human-annotated industrial test set against a Qwen2-VL-7B offline teacher, a six-layer Transformer deployed as the online student, and *GR + RelToken*, which predicts relevance from a post-SID relevance-token state via an MLP. As shown in Table 4, UniGD consistently outperforms both online alternatives across all metrics, demonstrating the advantage of explicit query–ad representation matching. With a 60M backbone and only 1.1M additional parameters, UniGD

Model Setting	Δ Revenue	Δ Impressions	Δ Imp. (LT)	Δ Imp. (New)	Δ CTR	CVR	Irrelevant Ratio	Latency
Baseline	–	–	–	–	–	–	6.79%	13.0 ms
GR+RelToken	–2.08%	+1.84%	–3.42%	–5.71%	–0.25%	–0.24%	17.30%	10.1 ms
UniGD-Gen	+3.54%	+2.61%	+10.40%	+15.30%	+0.56%	+0.41%	6.15%	12.9 ms
UniGD	**+5.78%**	**+3.12%**	**+16.80%**	**+24.60%**	**+0.94%**	**+0.68%**	**5.81%**	**8.7 ms**

Table 3: Online A/B results in the Kuaishou search advertising system. Δ denotes the relative change from the production baseline. **Imp. (LT)** and **Imp. (New)** denote impression changes for long-tail and new ads.

Model Setting	AUC	Spearman	Pearson
Offline Teacher	**0.907**	**0.805**	0.810
Online Student	0.881	0.783	0.784
GR+RelToken	0.890	0.795	0.796
UniGD	0.905	0.798	**0.811**

Table 4: Relevance estimation performance on our industrial human-annotated test set. The offline teacher is included as a reference and is not deployed for online inference.

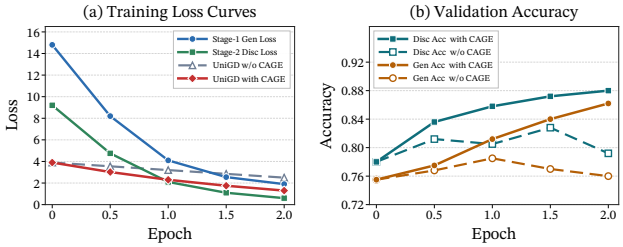


Figure 3: Training dynamics of UniGD. (a) Training loss across the three stages. (b) Generative and discriminative accuracy during joint training with and without CAGE.

also performs comparably to the offline teacher and achieves a marginally higher Pearson correlation. These results show that UniGD preserves strong relevance estimation within the generative retriever in production, avoiding a separately deployed online relevance model.

Ablation Study

We ablate the components of UniGD on NQ320K and our industrial advertising dataset. As shown in Table 2, we remove CAM and replace CAGE with standard joint optimization on both datasets. On our industrial dataset, we replace HAM with a shared codebook, SID space, and generation head across all material types. Removing CAM causes the largest degradation in Recall@1 and MRR@10, demonstrating the importance of codebook-anchored representations and explicit relevance estimation for candidate ranking. Removing CAGE consistently degrades all metrics, while replacing HAM with shared modeling reduces industrial performance, confirming the respective benefits of conflict-aware optimization and material-specific modeling. Figure 3 examines the effect of CAGE during joint optimization. With CAGE, both generative and discriminative validation accuracy improve steadily; without it, they decline after reaching intermediate peaks. These results suggest that CAGE mitigates optimization interference and stabilizes joint training.

Industrial Online Evaluation

We conduct a one-week controlled online A/B test in the Kuaishou search advertising system serving hundreds of millions of ad candidates, where newly indexed ads account for 36.9% of daily traffic. All groups share the same serving environment and differ only in their retrieval and relevance-estimation pipelines. We compare UniGD with the production cascade of GR and an external relevance model, GR + RelToken, and UniGD-Gen. UniGD-Gen retains UniGD's generative component but replaces its internal discriminative scoring with the production relevance model.

As shown in Table 3, UniGD improves revenue and new-ad impressions by 5.78% and 24.60%, respectively, over the production baseline, while reducing the Irrelevant Ratio. It also reduces retrieval latency from 13.00 ms to 8.70 ms by replacing the external relevance model with CAM lookup and lightweight internal scoring. GR + RelToken substantially increases the Irrelevant Ratio to 17.30%, indicating that a post-SID token state provides a less reliable relevance signal than explicit query–ad representation matching. UniGD's consistent advantages over GR + RelToken and UniGD-Gen across all major metrics demonstrate that directly integrating discriminative relevance estimation into generative retrieval improves both effectiveness and serving efficiency.

Conclusion

We present UniGD, a unified generative-discriminative framework for industrial search advertising. UniGD couples autoregressive candidate generation with explicit relevance estimation in a shared model, where CAM reconstructs ad representations from semantic codebooks, HAM accommodates heterogeneous ad materials, and CAGE coordinates the two objectives during joint optimization. Experiments on public benchmarks and large-scale industrial data demonstrate improvements in generative retrieval and relevance estimation, while online A/B tests in Kuaishou search advertising further validate its effectiveness in production. Beyond search advertising, UniGD can potentially be applied to recommendation and general information retrieval by coupling generative candidate retrieval with task-specific discriminative objectives, such as relevance estimation, preference prediction, and ranking. Evaluating UniGD in these broader domains is a promising direction for future work.

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